

Name: _____ Tutorial group: _____

Matriculation number:

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19 September 2022 MH1812 Test 1 60 minutes

QUESTION 1. **(30 marks)**

Use mathematical induction to show that

$$\frac{1}{2} \cdot \frac{4}{3} \cdot \frac{9}{4} \cdots \frac{n^2}{n+1} = \frac{n!}{n+1}.$$

whenever n is a positive integer.

[Proof:]

Base case: when $n = 1$, the left-hand side is $\frac{1}{2} = 1$ and the right-hand side is $\frac{1!}{1+1} = \frac{1}{2}$. The equality holds.

Inductive step: Suppose that the equality holds for $n = k$, and we shall show that it holds for $n = k + 1$, that is, we shall show that

$$\frac{1}{2} \cdot \frac{4}{3} \cdot \frac{9}{4} \cdots \frac{k^2}{k+1} \cdot \frac{(k+1)^2}{k+2} = \frac{(k+1)!}{k+2} \tag{1}$$

By induction hypothesis, the LHS of (1) equals

$$\frac{k!}{k+1} \cdot \frac{(k+1)^2}{k+2} = \frac{k! \cdot (k+1)}{k+2} = \frac{(k+1)!}{k+2},$$

which is exactly the RHS of (1).

Therefore, by mathematical induction, the original equality holds for all integers $n \geq 1$.

For graders only	Question	1	2(a)	2(b)	3(a)	3(b)	Bonus	Total
	Marks							

QUESTION 2. **(30 marks)**

(a) (10 points) Find $2022^{1812} \bmod 19$.

[Solution:] First, note that $2022 \bmod 19 = 8$, so $2022^{1812} \equiv 8^{1812} \pmod{19}$. Observe that $8^2 \equiv 7 \pmod{19}$, $8^3 \equiv 7 \cdot 8 \equiv -1 \pmod{19}$, then

$$8^{1812} \equiv (-1)^{1812/3} \equiv (-1)^{674} \equiv 1 \pmod{19},$$

that is, $2022^{1812} \bmod 19 = 1$.

[Solution 2:] Alternatively, we can continue from $8^3 \equiv 18 \pmod{19}$ and calculate $8^4 \equiv 18 \cdot 8 \equiv 11 \pmod{19}$, $8^5 \equiv 11 \cdot 8 \equiv 12 \pmod{19}$, $8^6 \equiv 12 \cdot 8 \equiv 1 \pmod{19}$. Notice that 1812 is divisible by 6, hence

$$8^{1812} \equiv 1^{1812/6} \equiv 1 \pmod{19}$$

and $2022^{1812} \bmod 19 = 1$.

(b) Let $\mathbb{R}^+$ denote the set of positive real numbers. For $x, y \in \mathbb{R}^+$, let $P(x, y)$ denote the predicate “ $x^4 + y^2 > 2.022x^2y$ ”. What are the truth values of these statements?

(i) (10 points) $\forall x \in \mathbb{R}^+, \exists y \in \mathbb{R}^+, P(x, y)$.

(ii) (10 points) $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}^+, P(x, y)$.

[Solution:]

(i) The statement is true. For any $x \in \mathbb{R}^+$, let $y = 2x^2$, then $x^4 + y^2 = 5x^4 > 4.044x^4 = 2.022x^2y$, i.e., $P(x, 2x^2)$ is true.

(ii) The statement is false. For any $x \in \mathbb{R}^+$, let $y = x^2$, then $x^4 + y^2 = 2x^4 < 2.022x^2y$, i.e., $P(x, x^2)$ is false.

QUESTION 3. (40 marks)

(a) (20 points) Determine whether $(\neg p \wedge (p \vee q)) \rightarrow q$ is a tautology.

[Proof:] It is a tautology. One can prove it by manipulating the formula or by a truth table.

$$\begin{aligned}
 & (\neg p \wedge (p \vee q)) \rightarrow q \\
 \equiv & ((\neg p \wedge p) \vee (\neg p \wedge q)) \rightarrow q \\
 \equiv & (F \vee (\neg p \wedge q)) \rightarrow q \\
 \equiv & (\neg p \wedge q) \rightarrow q \\
 \equiv & \neg(\neg p \wedge q) \vee q \\
 \equiv & p \vee \neg q \vee q \\
 \equiv & p \vee T \\
 \equiv & T
 \end{aligned}$$

p	q	$p \vee q$	$\neg p \wedge (p \vee q)$	$(\neg p \wedge (p \vee q)) \rightarrow q$
T	/	/	F	T
F	T	T	T	T
F	F	F	F	T

(‘/’ denotes an entry that need not be evaluated.)

(b) (20 points) Determine whether the following argument is valid. (Please state the names of the inference rules whenever you apply them.¹)

$$\begin{aligned}
 & q \rightarrow (r \rightarrow s); \\
 & t \rightarrow r; \\
 & p \vee q; \\
 & p; \\
 & \therefore \neg s \rightarrow \neg t.
 \end{aligned}$$

[Solution:] The argument is invalid. When $p = r = t = T$ and $q = s = F$, all premises are satisfied but the conclusion is not satisfied.

(This is the only counterexample. In a counterexample, s must be false and t must be true. The second premise is $t \rightarrow r$, thus r must be true. The fourth premise means that p must be true. The first premise then becomes $q \rightarrow (T \rightarrow F) \equiv q \rightarrow F$, for which to be true, it must hold that q is false.)

BONUS QUESTION. (10 marks)

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

What is the smallest positive integer n with n^2 ending in 0009? (This means that n^2 has at least five digits.) You must justify that the number you found is the smallest.

[Solution 1:] The assumption implies that $n^2 - 9$ is divisible by 10000, i.e., $(2^4 \cdot 5^4)|(n+3)(n-3)$. Since $n+3$ and $n-3$ cannot be both divisible by 5, it must hold that $5^4|(n+3)$ or $5^4|(n-3)$.

Suppose that $n+3 = 625k$, then $n-3 = 625k-6$, so $(n+3)(n-3)/5^4 = k(625k-6)$ is divisible by 2^4 . Note that $k(625k-6) \equiv k(k-6) \pmod{16}$, we can quickly check that the smallest desired $k = 6$. Thus $n = 625 \cdot 6 - 3$.

Suppose that $n-3 = 625k$, then $n+3 = 625k+6$. Similarly to the above, we have $k(k+6) \equiv 0 \pmod{16}$. The smallest k is thus $k = 2$, which means that $n = 625 \cdot 2 + 3 = 1253$.

Therefore, the smallest n comes from the second case and is $n = 1253$.

[Solution 2:] We first have $(2^4 \cdot 5^4)|(n+3)(n-3)$ as in the first solution. It is clear that n is odd, so $n = 2k+1$ for some k . This implies that $(5^4 \cdot 2^2)|(k+2)(k-1)$. Observe that $k+2$ and $k-1$ cannot be both divisible by 5, therefore, it holds either $5^4|k+2$ or $5^4|k-1$. To make n as small as possible, we try $k+2 = 5^4$ and $k-1 = 5^4$. When $k+2 = 5^4$, we have $k-1 = 622$, which is not divisible by 4 and so this is not a valid case. When $k-1 = 5^4$, we have $k+2 = 628$, which satisfies that $4|(k+2)$. Therefore, the smallest n corresponds to $k = 5^4 + 1 = 626$, that is, $n = 2k+1 = 1253$.

¹The inference rules you may use are: modus ponens, modus tollens, conjunctive simplification, conjunctive addition, disjunctive addition, disjunctive syllogism, hypothetical syllogism, disjunction elimination and rule of contradiction.